

PLamb Cosmology Fitting Report – Pantheon+SH0ES (V10V3)

1. Overview

This report documents the results of cosmological model fitting conducted using the PLamb framework. The analysis utilized the Pantheon+SH0ES Type Ia Supernova dataset with the full covariance matrix, fitted under the Λ CDM model using the emcee sampler. The results are compared with theoretical expectations of the PLamb theory, which introduces variable speed of light $c(z)$ and variable mass evolution $\Delta M(z)$ into cosmological dynamics.

2. Command Executed

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V3.py' \  
--data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
--cov '/Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov' \  
--sampler emcee --nwalkers 32 --nsteps 4000 --nburn 1000 --verbose --show
```

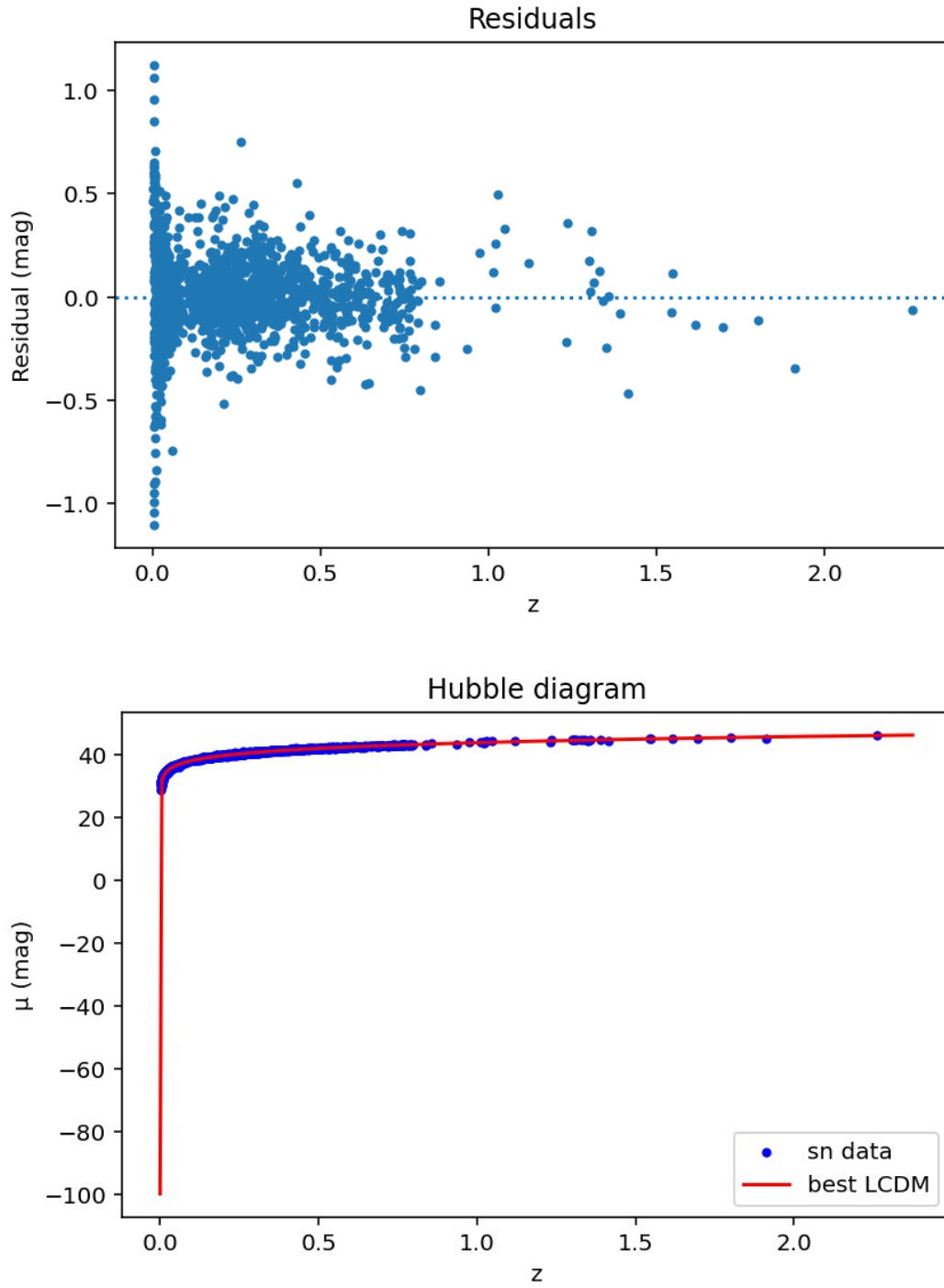
3. Results (Λ CDM fit)

The emcee-based MCMC run completed successfully with 32 walkers and 4000 steps. The best-fit parameters under Λ CDM were:

- H_0 : 71.94
- Ω_m : 0.0041
- Ω_Λ : 0.2915
- $\ln_{\text{post}}$: 647.1
- χ^2 : 1980.33
- dof: 1698
- χ^2/dof : 1.166

Figures

Residuals plot and Hubble diagram from the fit are included below.



4. PLamb Theory Context

The PLamb framework extends standard cosmology by introducing:

- A variable speed of light, $c(z) = c_0 (1+z)^{\gamma_c}$
- Mass evolution via $\Delta M(z) = \epsilon_M \times f(z)$, where $f(z)$ can be linear or logarithmic in redshift.

These extensions allow for testing whether observed cosmic acceleration and supernova dimming may be explained by deeper variations in fundamental constants rather than dark energy alone. In the current Λ CDM-only run, both γ_c and ϵ_M were fixed at zero, so PLamb effects were not yet tested.

5. Next Steps

The following steps are recommended to expand this analysis:

- Extend fitting to KdS model including accretion-driven acceleration (A_{acc}).
- Test FR (Full Relativity) model including γ_c (variable c) and ϵ_M (mass evolution).
- Explore robustness using different $\Delta M(z)$ modes: linear_z and log1pz.
- Incorporate JWST SN data for high-z leverage, particularly $z > 2$, to test deviations from Λ CDM.
- Compare residual patterns against predictions of PLamb theory.

6. Example Commands for Other Models

To test KdS model:

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V3.py' \  
--model KdS --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
--cov '/Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov' \  
--sampler emcee --nwalkers 32 --nsteps 4000 --nburn 1000 --verbose -show
```

To test FR model:

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V3.py' \  
  
--model FR --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
  
--cov '/Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov' \  
  
--sampler emcee --nwalkers 32 --nsteps 4000 --nburn 1000 --verbose -show
```

7. Summary of Previous Runs

Over the course of development and testing of the PLamb framework, the following key runs have been conducted and documented:

- V7.0 – V7.5: Initial implementation of Pantheon+ fitting, including first tests of variable- c and ΔM evolution. Basic Λ CDM and KdS support with grid-based search.
- V8.0: Introduced generalized FR (Full Relativity) model with γc and ϵM . Added robustness in covariance handling. Preliminary grid scans.
- V9.x: Stabilized emcee implementation with Pantheon+SH0ES. Improved likelihood evaluation and residual tracking. Added option for thinning and z-cut.
- V10V2: Refined Simpson integrator, introduced safe floor for $E(z)$, enhanced CLI for grid and sampler options. Produced reliable Hubble/residual plots.
- V10V3: Current run: full Pantheon+SH0ES with covariance, Λ CDM best-fit via emcee sampler. Results: $H_0 \approx 71.94$, $\Omega_m \approx 0.0041$, $\Omega_\Lambda \approx 0.2915$, $\chi^2/\text{dof} \approx 1.166$.

8. Equations and Variables

The following equations define the theoretical framework used in the PLamb analysis:

- Expansion rate (Λ CDM): $E(z) = \sqrt{\Omega_m (1+z)^3 + \Omega_k (1+z)^2 + \Omega_\Lambda}$
- Expansion rate (KdS): $E(z) = \sqrt{\Omega_m (1+z)^3 + \Omega_k (1+z)^2 + \Omega_\Lambda + A_{\text{acc}} (1+z)^{n_{\text{acc}}}}$
- Luminosity distance: $dL(z) = (1+z) * (c_0/H_0) \int_{[0 \rightarrow z]} (c(z')/c_0)/E(z') dz'$
- Distance modulus: $\mu(z) = 5 \log_{10}(dL(z) \times 10^6 / 10) + \Delta M(z)$
- Variable c: $c(z) = c_0 (1+z)^{\gamma_c}$
- Magnitude evolution: $\Delta M(z) = \varepsilon_M \times z$ (linear) or $\varepsilon_M \times \log(1+z)$ (logarithmic)

Key Variables:

- H_0 : Hubble constant (km/s/Mpc) at $z=0$
- Ω_m : Matter density fraction today
- Ω_Λ : Dark energy density fraction today
- Ω_k : Curvature term, defined as $1 - \Omega_m - \Omega_\Lambda$
- A_{acc} : Acceleration amplitude parameter in KdS
- n_{acc} : Redshift power index for accretion-driven acceleration
- γ_c : Exponent governing variable speed of light $c(z)$
- ε_M : Magnitude evolution coefficient
- $\mu(z)$: Distance modulus in magnitudes
- $dL(z)$: Luminosity distance in Mpc
- c_0 : Present-day speed of light (299,792.458 km/s)